AP CALCULUS AB - UNIT 1

Guided Notes 1.13: Removing Discontinuities

Name: ____________ **Class:** ____________ **Date:** ____________

Learning Objectives and Essential Question

Essential question: How can we decide whether a discontinuity can be removed by changing a point value?

- Decide whether a discontinuity can be removed by changing a point value.
- Find parameters that make piecewise functions continuous at joins.
- Create or cancel common factors to identify removable holes.
- Solve multi-boundary continuity systems and justify uniqueness or impossibility.

Key Knowledge, Vocabulary, and Notation

Repair value	If $\lim_{x\to c} f(x) = L$ is finite, define or redefine $f(c) = L$.
Join condition	At a piecewise boundary require left limit = right limit = assigned value.
Factor creation	A parameter may be chosen so the numerator shares a denominator factor.
Multiplicity	Cancellation repairs a hole only when every problematic denominator factor is removed.
Possible outcomes	A repair problem can have a unique solution, infinitely many solutions, no solution, or require changing more than a point value.

Concept Development and Strategy

1. First classify the discontinuity. A point-value repair is possible only when a finite common limit already exists.
2. For piecewise joins, write equations before solving parameters.
3. For rational expressions, use the factor theorem: $x - c$ divides a polynomial $P(x)$ exactly when $P(c) = 0$.

Worked Example: Fill a hole

Define $f(3)$ so that $f(x) = \frac{x^2-9}{x-3}$ is continuous at 3.

Answer and Reasoning

For $x \neq 3$, $f(x) = x + 3$, so the limit is 6. Define $f(3) = 6$.

Worked Example: One parameter at a join

Find k so $f(x) = kx + 1$ for $x < 2$ and $f(x) = x^2 - 1$ for $x \geq 2$ is continuous at 2.

Answer and Reasoning

Require $2k + 1 = 3$, giving $k = 1$.

Worked Example: Two joins

Find a, b so $f(x) = x + 1$ for $x < 0$, $f(x) = ax + b$ for $0 \leq x < 2$, and $f(x) = 5$ for $x \geq 2$ is continuous.

Answer and Reasoning

At 0, $b = 1$. At 2, $2a + b = 5$, so $a = 2$. Thus $(a, b) = (2, 1)$.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. What value repairs $f(x) = \frac{x^2-4}{x-2}$ at $x = 2$?

2. What value repairs $g(x) = \frac{x^2+x-6}{x-2}$ at $x = 2$?

3. Find k so $f(x) = \begin{cases} 2x + k, & x < 1 \\ x^2 + 3, & x \geq 1 \end{cases}$ is continuous at 1.

4. Find a so $\frac{x^2+ax-6}{x-2}$ has a removable discontinuity at 2. Then find the repair value.

5. Find a, b so $f(x) = \begin{cases} ax + b, & x < 1 \\ x^2, & 1 \leq x < 3 \\ 2x + b, & x \geq 3 \end{cases}$ is continuous at both joins.

6. Find all a such that $\frac{x^2+(a-3)x-3a}{(x-3)^2}$ has a removable discontinuity at 3.

7. For positive integer n, choose coefficients so $\frac{x^n - a^n}{x - a}$ extends continuously to $x = a$, and give the extension value.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. To repair a removable discontinuity at $x = c$, define $f(c)$ to be

(A) 0
(B) the left-hand limit only
(C) the finite two-sided limit
(D) any real number

2. For $\frac{x^2+kx-12}{x-3}$ to have a removable discontinuity at 3, k must be

(A) -7
(B) -1
(C) 1
(D) 7

3. A calculator estimates a common limit of 4.999999 at $x = 2$ while $f(2)$ is undefined. The natural exact repair conjecture is

(A) 0
(B) 2
(C) 5
(D) $+\infty$

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Construct a one-parameter family of distinct functions, each with a removable discontinuity at $x = 2$ and repair value 7.